Materials Resource Management

Universität Augsburg University

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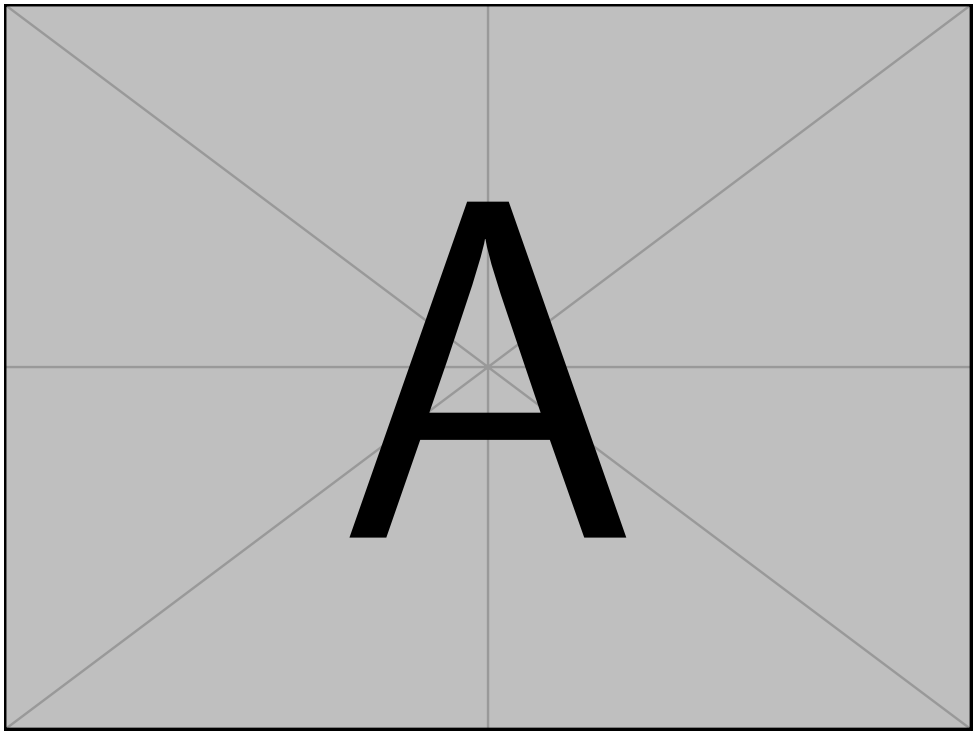
The English Title

The author



Supervisor: The supervisor

First reviewer: The first reviewer

Second reviewer: The second reviewer

Augsburg, April 2023

Materials Resource Management

Universität Augsburg University

Author information

The author

Matr.-Nr.: The registration number

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The author declares to have prepared this thesis independently and only with the sources and aids indicated in the bibliography.

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Materials Resource Management

Universität Augsburg University

Task

for

The author

(Matr.-Nr.: The registration number)

The English Title

Der deutsche Titel

A concise description of the task given to the student submitting this thesis.

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Acknowledgments

This is an opportunity to thank people supporting your thesis, e.g. technicians, supervisors, friends providing mental support, etc.

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Abstract

An abstract is a condensation of the content into one paragraph, at most one page. The abstract should stand on its own and present the motivation, methods, goals, and results of the work to someone who does not have time to read the entire work. It must enable a reader to decide whether he should read this work. The abstract is of great importance and should be formulated very consciously, because crucial key points of the work must become clear.

Context: First, the reader should get on board. For this purpose, one or two sentences are used to introduce the subject.

State of research: At this point, the current state of research is summarized very briefly. In particular, shortcomings of previous results should be addressed in order to work towards one's own research question.

Research gap: Here the own research question is worked out. What is missing so far or which aspects need to be examined more closely?

Own approach: This section describes the methods used - what is one's approach to solving the research question?

Results: What came out of the work and which conclusions may be drawn?

You may also summarize your work in a graphical abstract and use it as the title image of this thesis.

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Acronyms

CMC	Ceramic Matrix Composites.
SVD	Singular Value Decomposition.

Symbols

Symbol	Description	Unit
$\mathbb{C}$	Stiffness tensor	GPa
π	Geometrical constant for circles	-

1 Introduction

This chapter should start with the motivation outlining the necessity and novelty of the scientific endeavor. The goal of the introduction is the elaboration of the research question from which the objective of the thesis is derived. The specific aspects which are examined are to be illustrated in a brief and concise manner. This chapter, as well as the abstract, can be finalized towards the end of the thesis in order to ensure the coherence with the last chapter (conclusion).

2 Fundamentals and state of research

This chapter is supposed to prepare the reader for the following chapters. It includes the analysis of relevant scientific work on the given topic (state of the art) as well as the elaboration of the fundamental knowledge necessary for the presentation of one's own work. The reader can be assumed to have a professional background yet without familiarity with the specific field of research. The topic of this thesis is to be contextualized within current research activities and distinguished from similar research.

2.1 Citations

This chapter will make use of many citations, hence some notes on citing are summarized here. There are two options to cite in LaTeX:

1. The command `\cite{key}` creates an automatically updated reference number, such as in this example: *This is a sample citation [1].*
2. The command `\citet{key}` adds the author name, such as in this example: *Einstein [1] wrote an important paper*. One author or two authors should be explicitly named, more authors are denoted *Einstein et al.* . The command `\citet{key}` takes care of this automatically.

The keys and bibliographic data for the citations are listed in the file `literature.bib`. This can be created manually using the *BibTex* format or by any literature management software. Usage of such a literature management software (e.g. Citavi, Mendely, ResearchRabbit, or Endnote) is highly recommended. The bibliography should contain just references, which are actually used and referred to. In the default configuration of this template, the template takes care of this automatically.

Images, tables, technical specifications, etc. have to be cited as well. Adaptions of a figure from literature should be indicated as well, e.g. by writing *adapted from ....*

2.2 Language and style

A scientific thesis requires concise and precise language. Using past tense is reasonable in this section, as other people may have contributed to the state of research in the past, but apart from this, present tense should be used throughout the work. Expressions like *high temperature*, *good rheology* or every-day language should be avoided.

Use sections and subsections reasonable and consistent. It should not be too granular and subsections do not make any sense, if there is not at least one other subsection at the same level. It is an important task of a student thesis to find a good structure and should be discussed in close coordination with supervisors.

3 Methodology

This chapter should describe the applied methods or the newly developed methods to obtain research results. There is a difference between an experimental thesis and a modeling thesis in this chapter. The first one will have a rather descriptive character, while the second one will include more own developments in this chapter.

3.1 Figures

This chapter will probably make use of several figures, e.g. showing experimental setups or sketches. The subsequent code showcases insertion of an image, where the expression *path* should be replaced with the actual path to your image.

```
\begin{figure}[ht]
    \centering
    \includegraphics[width=0.75\textwidth]{path}
    \caption{A caption describing the image.}
    \label{fig:image_label_a}
\end{figure}
```

This sample code results in the image shown in Figure 3.1. Please note that references to figures are capitalized, e.g. `Figure \ref{fig:image_label_a}`. Every image must be referenced in the text, there is no point having an image that is never mentioned in the text. Preferable, you should refer to the image before it appears in the document.

Make sure that the image is readable, i.e. features are large enough and the image quality is high. For pictures, use `JPEG` format, for diagrams and sketches use `PNG` format and for diagrams its best to use `PDF` format for optimal trade-offs between file size and quality. For diagrams, make sure to include axis labels with a sufficiently large font and use *in <unit>* instead of rectangular brackets to indicate units. Make sure that colors are easy to distinguish, some background on colors may be found https://colorbrewer2.org and https://bids.github.io/colormap/.

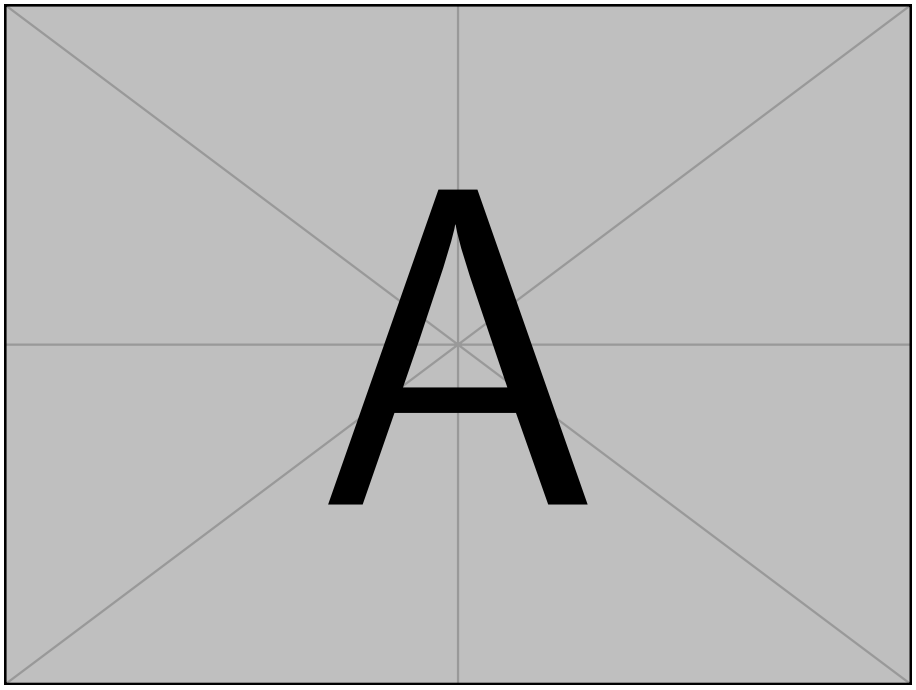


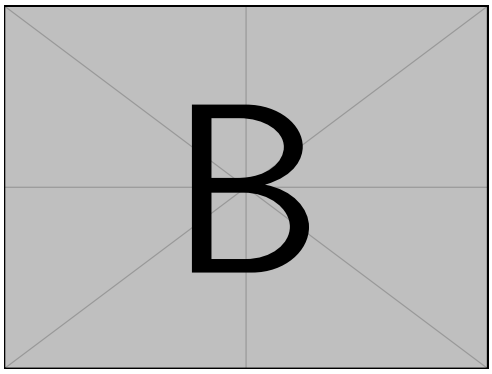
Figure 3.1: A caption describing the image. The caption should stand on its own and include a reference, if the image is adapted from another author.

We may also combine multiple images, for example by the following code

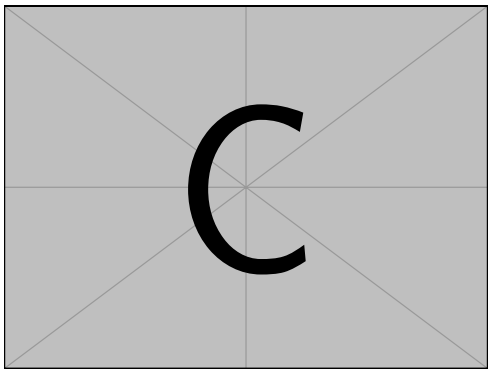
```
\begin{figure}[ht]
 \centering
 \begin{subfigure}[b]{0.4\textwidth}
     \centering
     \includegraphics[width=\textwidth]{example-image-b}
     \caption{Sample image B}
     \label{fig:y equals x}
 \end{subfigure}
 \hfill
 \begin{subfigure}[b]{0.4\textwidth}
     \centering
     \includegraphics[width=\textwidth]{example-image-c}
     \caption{Sample image B}
     \label{fig:three sin x}
 \end{subfigure}
\caption{Two images side by side}
\label{fig:two_figs}
```

```
\end{figure}
```

which results in Figure 3.2. This is useful for similar images or images comparing similar information.



(a) Sample image B



(b) Sample image B

Figure 3.2: Two images side by side

In diagrams with nominal features, i.e. they do not have a natural order such as *bananas*, *apples* and *peaches*, bar charts should be used. Never interpolate such data with any lines. If data is obtained from experiments, information about scatter must be provided, e.g. with box plots showing percentiles or at least by providing a standard deviation for Gaussian processes.

3.2 Tables

Tables are included inside via the environment `table`. There are several ways to create the actual table. The most common are the `tabular*` and `tabularx` environments. An example of a table can be found below in Table 3.1.

Table 3.1: Headline of the example table

	Thickness t in mm	**Young's modulus** E in GPa	**Strength** R_m in MPa	**Density** ϱ in $\mathrm{kg\,m^{-3}}$
Material 1	100.0	2000	1500	3.0
Material 2	2.0	3000	880	55.0
Material 3	0.6	4	713	52.8

Tables must have headlines and must be referred to in the text. Numbers should be aligned to the right and vertical lines should be omitted.

3.3 Equations

Important formulas should be inserted via the `equation` environment, which numbers them and offers the option to refer to equations via `\eqref`. For example, the code

```
\begin{equation}
    f(x)  = \sin \left(\frac{\sin(x)}{x} \right)
        \label{eq:key}
\end{equation}
```

results in

$$f(x) = \sin \left(\frac{\sin(x)}{x} \right) . \tag{3.1}$$

Note that equations are part of the text and proper punctuation is needed. Specific operators and functions like $\ln$, $\cos$ and the d in differentials should be non-italic.

You may also use `$ 1/x $` for short in-line equations in the text like $1/x$. This is a reference to Equation 3.1.

3.4 Units

For dealing with dimensional values, the `siunitx` package is useful. In text or math mode a value can be assigned a unit with the following syntax:

```
\SI{value}{units}
```

An example is

```
\SI{2.0}{\micro\gram\per\cubic\metre}
```

resulting in the expression $2.0\,\mu\mathrm{g}\,\mathrm{m}^{-3}$. The package takes care of proper spacing between numbers and units as well as line breaks.

4 Results and discussion

4.1 First aspect

4.1.1 Description

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning.

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4.1.2 Results

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"?

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4.1.3 Discussion

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should be written in of the original language. There is no need for special content, but the length of words should match the language.

4.2 Second aspect

4.2.1 Description

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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4.2.2 Results

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4.2.3 Discussion

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4.3 Third aspect

4.3.1 Description

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4.3.2 Results

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

4.3.3 Discussion

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is

there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

5 Conclusion and outlook

This is a short chapter summarizing the work that has been performed and conclusion to be drawn from it. In addition, it should provide an outlook on further steps in the research field. The final part of the thesis should form a unit with the introduction and pick up on issues already highlighted there. Together they form one of the most important parts of a good thesis as they motivate the reader to carefully study the main part of one's work.

Bibliography

[1] Albert Einstein. Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig? *Annalen der Physik*, 323(13):639–641, 1905. DOI: 10.1002/andp.19053231314.